Draft Labels and Career Opportunities: Regression Discontinuity Evidence from the NBA Draft

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Abstract

What is the causal effect of being selected in the first round of the NBA Draft, rather than the second round, on a player's NBA outcomes? I use a sharp regression discontinuity design around the year-specific first-round cutoff and a player-level dataset built from Basketball-Reference covering the first 60 picks of every NBA Draft from 1980 to 2023 (2,578 players). Players selected just inside the first round play approximately 25 more games over their first three seasons, roughly 1.3 more NBA seasons, and about 84 more career games than players selected just outside the cutoff. First-three-year salary shows essentially no jump at the cutoff, and placebo cutoffs within each round produce small and statistically insignificant estimates. The evidence points to a broader opportunity channel: first-round status appears to increase roster security, organizational investment, and early playing opportunities, which then accumulate into larger long-run differences in career outcomes.

1 Introduction

In the 2014 NBA (National Basketball Association) Draft[1], Kyle Anderson was selected with the 30th pick, the final pick of the first round. Damien Inglis was selected with the 31st pick, the first pick of the second round. They were separated by only one draft slot, but their NBA careers ended up looking very different. Anderson became a long-term rotation player, while Inglis played only briefly in the NBA. This one comparison does not prove that the first-round label caused the difference. But it shows why the cutoff is interesting: two players can be drafted almost next to each other, but the first-round label may lead to very different expectations and opportunities.

Motivated by this example and many similar cases, this paper asks the following research question: What is the causal effect of being selected in the first round of the NBA Draft, rather than the second round, on a player's NBA outcomes? A simple comparison between all first-round and second-round players would not answer this question because first-round players are usually better prospects. If first-round players have better careers on average, that may simply reflect higher ability before entering the league. To deal with this problem, I use a sharp regression discontinuity design (RDD) around the cutoff between the first and second rounds. The design compares players drafted just inside and just outside the first round, where players should be relatively similar in ability but receive drastically different draft labels.

This question is worth studying because it speaks to a broader issue beyond basketball or sports. In many labor markets, people are sorted into categories early in their careers, and those categories can affect how much opportunity, patience, mentorship, and investment they receive. The NBA Draft is a useful setting to study this question because the first-round cutoff is clearly defined, difficult for individual players to manipulate, and linked to career outcomes that can be measured over time. A player selected at the end of the first round may receive more contract security, more development opportunities, and more organizational investment than a player selected one pick later purely because of their first-round status. A discontinuity at this boundary would suggest that first-round status itself has a significant effect on player outcomes, beyond the small difference in draft position among these players.

In order to implement the RDD research design, I built a player-level dataset from Basketball-Reference covering the first 60 picks of every NBA Draft from 1980 to 2023. The dataset includes draft position, player background variables, season-by-season NBA statistics, career outcomes, college statistics, and salary data. Since the number of first-round picks changes over time, I define the cutoff separately for each draft year. The running variable is each player's pick number relative to that year's first-round cutoff, and the main specification estimates the discontinuous jump in outcomes at the cutoff.

[1] Appendix A provides a brief overview of the NBA Draft mechanism.

The main result is that players selected just inside the first round experience substantially better NBA outcomes than players selected just outside the first round. The strongest effects appear in opportunity-based outcomes, especially games played, minutes played, career length, and cumulative production. In contrast, the discontinuity in first-three-year salary is small and statistically insignificant, suggesting that the first-round advantage does not operate mainly through higher initial pay. Instead, the evidence points to a broader opportunity channel: first-round status appears to increase roster security, organizational investment, and early playing opportunities, which then accumulate into larger long-run differences in career outcomes.

This paper contributes to existing literature, particularly Zhou, Hu, and Shi's 2023 paper, by expanding the sample across more draft eras, separating early-career outcomes from long-run outcomes, using broader performance measures such as win shares, VORP, and box plus/minus, and tracing whether the first-round gap persists over the career. Overall, the results suggest that the first-round label is not merely a reflection of how teams valued a player before the draft. It can also shape the opportunities a player receives after entering the league, affecting his ability to build a sustained NBA career.

2 Literature Review and Extensions

A growing literature studies how draft position affects player outcomes in professional sports, with a central question being whether higher draft status reflects true differences in player ability or instead generates advantages through organizational behavior. This distinction is important for this paper, which examines whether being selected in the first round of the NBA Draft has a causal effect on player outcomes.

Early work in this area focuses on the relationship between draft position and performance, but largely treats draft order as a proxy for ability. For example, Watave (2016) shows that while higher draft picks tend to perform better on average, the relationship is noisy, and players selected close to each other in the draft often have similar outcomes. Similarly, Greene (2015) finds that although college statistics can predict some aspects of NBA performance, a large portion of variation remains unexplained. These findings suggest that draft position is an imperfect measure of underlying talent, particularly for players selected near cutoff points. This observation is important because it supports the idea that players just inside and just outside the first round may be comparable in ability.

In addition to these descriptive relationships, evidence from applied analyses highlights discontinuities in outcomes at key draft thresholds. A widely cited analysis from Harvard Sports Analysis Collective (2012) compares players selected at the end of the first round with those at the beginning of the second round and finds large differences in career outcomes, despite minimal differences in draft position. This pattern is difficult to explain by differences in talent alone and instead suggests that crossing the first-round threshold may itself generate advantages.

There is also related evidence from the broader sports economics literature beyond basketball. Keefer (2017) studies the NFL (National Football League) draft using a regression discontinuity design around the first-round cutoff and finds that first-round players receive more playing time but are not significantly more productive than those just outside the cutoff. He interprets this as evidence of organizational investment and sunk-cost behavior, where teams allocate more opportunities to players in whom they have made larger initial investments. This provides a potential mechanism through which draft status can affect outcomes independently of ability, suggesting that any discontinuity observed at the NBA first-round cutoff may reflect differences in organizational treatment rather than differences in underlying talent.

The existing literature most closely related to this study is Zhou, Hu, and Shi (2023), which applies a regression discontinuity design to NBA draft data from 1995 to 2019. Comparing players selected just inside and just outside the first round, they find large discontinuities in salary, points, games, and minutes played: a first-round pick earns about 163% more over his career and scores about 167% more points than a comparable second-round pick. They argue that these gaps do not reflect differences in ability, which is smooth at the cutoff, but instead arise from features of the rookie contract system introduced under the 1995 Collective Bargaining Agreement (CBA), including longer contract length and a greater number of guaranteed years. These features increase teams' incentives to invest in first-round picks and raise the sunk cost of not playing them. Zhou et al. support this interpretation using a mediation specification and a difference-in-discontinuities design around the 2005 CBA, which reduced guaranteed years and attenuated the round-1 premium.

This paper builds on Zhou et al. by adopting the same regression discontinuity framework and extending the analysis along four dimensions. First, the sample is expanded from 1995–2019 to 1980–2023. The pre-1995 period predates the modern rookie wage scale, when contract differences at the first-round cutoff were less structured, while the post-1995 period spans multiple CBA regimes, including the 1995 introduction of the rookie scale, the 2005 reform, and the 2017 increase in rookie pay. Estimating the discontinuity separately across these regimes provides a direct test of whether the first-round premium tracks institutional changes in contract rules.

Second, the paper separates early-career outcomes from long-run outcomes. Zhou et al. estimate the effect of first-round selection using outcomes such as log salary and log performance measures, which capture both whether a player makes it to the NBA and how much he produces once he does. In contrast, in this paper, I estimate these effects separately for NBA entry, salary in the first three seasons, and long-run career outcomes. The estimated jump in first-three-year salary is small and not statistically significant, consistent with the rookie scale compressing initial pay across the cutoff. In contrast, larger differences emerge in long-run outcomes such as career length, games played, minutes played, points, and total salary. The value-based measures, including win shares and VORP, are generally positive but less precisely estimated. This pattern suggests that the first-round advantage does not come from higher initial pay, but from sustained differences in opportunities that build over time.

Third, the paper incorporates a broader set of player-value metrics and integrates them into a more systematic analysis of the first-round effect. While Zhou et al. include some per-minute measures, their analysis focuses primarily on cumulative outcomes such as total points and total salary. This paper extends the analysis by incorporating more comprehensive value-based metrics, including win shares[2], value over replacement player (VORP)[3], and box plus/minus[4], alongside efficiency measures. By examining these outcomes together, the paper provides a more complete picture of how the first-round effect operates across different dimensions of performance. In particular, comparing value-based and cumulative measures helps assess whether the discontinuity reflects differences in productivity or differences in opportunities over a player's career.

Fourth, the paper traces the round-1 effect over the career rather than focusing only on cumulative outcomes. This provides a direct test of persistence. The results show that the gap in games, minutes, and production emerges early and remains positive throughout players' careers, rather than disappearing once rookie contracts expire. This pattern is inconsistent with a purely mechanical contract effect and instead suggests that early differences in opportunity translate into lasting differences in career trajectories.

Together, these extensions use variation across CBA regimes, outcome measures, and career stages to test whether the first-round discontinuity is driven by contract rules, organizational investment and reputation, or differences in skill development induced by early opportunities.

3 Data

This paper uses a player-level NBA draft dataset that I scraped from Basketball-Reference (https://www.basketball-reference.com/draft/). The dataset covers the first 60 picks of all the NBA drafts from 1980 to 2023 and combines draft information, player background variables, season-by-season NBA statistics, career statistics, college statistics, as well as salary information. I built the dataset around the main research design of the paper, which is to compare players selected just

[2]Win shares: An advanced basketball statistic that estimates the number of team wins attributable to a player's offensive and defensive contributions.

[3]Value over replacement player (VORP): An advanced basketball statistic that estimates a player's total value relative to a replacement-level player.

[4]Box plus/minus: An advanced basketball statistic that estimates a player's per-possession impact relative to league average, based on box-score performance.

inside and just outside the first round using an RDD design.

The main draft variables are draft year, overall pick, draft round, drafting team, player name, college, and whether the player eventually played in the NBA. Since the number of first-round picks changed over time, I do not use a fixed cutoff such as pick 30 for every year. Instead, I define the first-round cutoff separately for each draft year. The variable *round1_cutoff* equals the last pick of the first round in that draft. I then define the running variable as the player's overall pick minus the first-round cutoff in that year. With this definition, a value of 0 means the player was the last pick of the first round. Negative values are players selected before the cutoff, and positive values are players selected after the cutoff. I also define *D_round1*, which equals 1 if the player was selected in the first round and 0 otherwise. This is the main treatment variable in the regression discontinuity design.

The outcome variables include both season-level outcomes and career-level outcomes. I obtained a full season-by-season dataset for each drafted player's NBA career, which allows me to observe how outcomes evolve over time after the draft. The season-level outcomes include games played, minutes played, points, rebounds, assists, win shares, VORP, and salary. This structure allows me to study early-career outcomes, especially the first three seasons, while also keeping the flexibility to examine later seasons. I focus especially on the first three years because this is the period when draft status is most likely to affect players' development through playing time, roster security, and team investment. If teams treat first-round players differently, that should show up early through more minutes, more patience, or more chances to develop.

I also construct the same types of outcomes at the career level, which allows me to examine whether early differences persist into longer-run NBA success and total career earnings. However, it is worth noting that career-level outcomes should be interpreted more carefully because, compared to early-career outcomes, they are more heavily affected by later factors after the draft, such as injuries, trades, player development, team fit, and changes in team situations.

One important data choice is that I keep every drafted player in both the season-by-season dataset and the career-level dataset. For the season-by-season dataset, players are observed for each NBA season they played. Players who never appeared in the NBA are still retained in the dataset with their NBA performance outcomes coded as zero. For the career-level dataset, players who never appeared in the NBA are also retained, with their career performance outcomes coded as zero. I make this choice because not playing is itself an outcome that contains valuable information about a player's ability, opportunities, or both. Dropping these players would remove an important part of the story, especially if second-round players are statistically less likely to receive NBA opportunities in the first place.

The dataset also includes background and pre-draft variables, such as height, weight, position, age at draft, birth date, and college statistics when available. These variables summarize player characteristics and pre-draft production, including physical profile, position, playing time, scoring, shooting efficiency, and other box-score performance measures. They help describe the types of players being compared around the cutoff and can also be used as controls in some specifications for robustness checks. In addition, they are useful for checking whether players near the cutoff appear similar before entering the NBA. This is important because the validity of the RDD design relies on the assumption that underlying player ability changes smoothly around the cutoff, which will be discussed in greater detail in the empirical strategy section.

The main strength of the dataset is that it fits the research design closely. The analysis requires a dataset that identifies each player's draft position, the relevant first-round cutoff for that year, and later NBA outcomes. This comprehensive dataset provides all of those pieces. It also allows me to separate short-run outcomes from long-run outcomes. The early-career outcomes are useful for studying early opportunity and team investment, while the cumulative career outcomes are useful for studying whether those early differences persist over time.

There are also limitations. Basketball-Reference provides detailed public data, but it is not the same as an ideal scouting dataset. Ideally, I would observe teams' private draft boards, workout results, medical reports, interview evaluations, and internal scouting grades. Since these variables are not available, I cannot directly show that teams viewed players around the cutoff as identical prospects. Instead, the RDD relies on the assumption that underlying player ability changes smoothly around the cutoff. To support this assumption, I use available background and college statistics to check whether players just before and just after the cutoff look similar before entering the NBA. However, these pre-draft comparisons are limited by the fact that college statistics are missing for some international players, high school players, and players from less traditional pre-draft paths. Additionally, salary information is less complete for some older draft years, and career outcomes for recent draft classes are not fully observed. For this reason, I interpret salary and career-level results carefully and restrict some career analyses to specific draft-year ranges as a robustness check.

Overall, despite these limitations, the dataset is appropriate for this paper because it contains the key variables needed for the analysis: draft status, the year-specific first-round cutoff, the running variable, player background characteristics, early NBA outcomes, career NBA outcomes, and salary outcomes. Table 1 summarizes the main variables included. The sample includes 2,578 drafted players from 44 draft years, with substantial variation in both early-career and career outcomes, which is useful for studying whether small differences around the first-round cutoff translate into larger differences over time.

Table 1. Summary statistics

Variable	N	Mean	SD	Min	Median	Max
Panel A. Sample composition						
Drafted players, 1980–2023	2,578					
Draft years	44					
First-round picks (share)	1,232	0.478				
Players who appeared in NBA (share)	2,201	0.854				
Players with NCAA stats on record (share)	2,194	0.851				
Panel B. Pre-draft characteristics						
Age at draft (yrs)	2,555	21.68	1.38	17.66	21.95	27.65
Height (in.)	2,561	79.05	3.46	63.00	79.00	91.00
Weight (lbs)	2,561	217.4	26.9	136.0	215.0	403.0
Position (1=PG … 5=C)	2,462	3.07	1.40	1.00	3.00	5.00
College games played (last NCAA yr)	2,194	31.6	4.8	3.0	32.0	41.0
College MPG (last NCAA yr)	2,194	31.3	6.3	0.0	32.5	40.0
College PPG (last NCAA yr)	2,194	16.61	4.61	0.80	16.50	36.30
College FG% (last NCAA yr)	2,194	50.43	6.47	0.00	50.00	76.90
Panel C. Early-career outcomes (sums over first 3 NBA seasons)						
Games (yrs 1–3)	2,578	119.6	84.6	0.0	136.0	247.0
Minutes (yrs 1–3)	2,578	2,562	2,468	0	1,956	9,871
Points (yrs 1–3)	2,578	1,069	1,192	0	674	6,585
Win shares (yrs 1–3)	2,578	4.51	6.12	−2.70	1.90	46.00
VORP (yrs 1–3)	2,578	0.75	2.43	−5.50	0.00	22.40
Salary, first 3 yrs ($M)	2,578	3.43	4.37	0.00	2.11	34.82
Played in NBA in season 1 (0/1)	2,578	0.854	0.353	0.000	1.000	1.000
Panel D. Career outcomes (cumulative through 2024 season)						
Career length (yrs)	2,578	5.86	5.05	0.00	4.00	23.00
Career games	2,578	327	344	0	200	1,622
Career minutes	2,578	7,950	10,214	0	3,098	61,030
Career points	2,578	3,470	5,100	0	1,128	43,440
Career win shares	2,578	17.18	29.24	−2.70	3.20	276.80
Career VORP	2,578	4.33	11.96	−8.50	0.00	159.40
Total career salary ($M)	2,564	26.93	52.29	0.00	4.92	531.32

Notes: NBA drafts 1980–2023 (44 cohorts, 60 picks per year from 1989 onward; fewer in early years). Pre-draft characteristics merge Basketball-Reference player profiles and the last NCAA season on record. Early-career outcomes sum over a player's first three NBA seasons; players who never appear in the NBA contribute zeros for counting outcomes (intent-to-treat). Career outcomes cumulate through the 2024 season and are right-censored for the 2017–2023 cohorts. Salary is expressed in millions of current U.S. dollars and is reported as zero for ∼20% of picks (mostly never-NBA or pre-1985 players).

4 Empirical Strategy

The main empirical challenge in this paper is that draft status is not randomly assigned. Players selected in the first round are generally viewed as better prospects than players selected in the second round. They may have better pre-draft production, stronger physical profiles, more upside, or better information from scouting reports that I cannot fully observe in my data. For this reason, a simple comparison between first-round and second-round players would not identify the causal effect of first-round status. In general, if first-round players have better NBA outcomes, that difference could simply reflect the fact that they were better players before entering the league.

To address this selection problem, I use a regression discontinuity design around the cutoff between the first and second rounds of the NBA Draft. The design compares players selected just inside the first round with players selected just outside the first round. These players are very close in draft position and are likely to be similar in underlying ability, but they receive different draft labels. The player selected with the last pick of the first round receives first-round status, while the player selected with the next pick begins the second round. Therefore, if there is a discontinuous jump in outcomes exactly at this cutoff, it is more likely to reflect the effect of first-round status rather than smooth differences in player ability.

As defined in the data section, the running variable is the player's overall pick relative to the year-specific first-round cutoff. Formally, I define:

$$c_{it} = overall\_pick_{it} - round1\_cutoff_t \tag{1}$$

where $overall\_pick_{it}$ is player i's overall pick number in draft year t, and $round1\_cutoff_t$ is the final pick of the first round in that draft year. With this definition, $c_{it} = 0$ means that the player was the last pick of the first round, negative values represent players selected before the cutoff, and positive values represent players selected after the cutoff. The treatment variable is:

$$D_{it} = \mathbf{1}\{c_{it} \leq 0\} \tag{2}$$

where $D_{it} = 1$ if the player was selected in the first round and $D_{it} = 0$ otherwise. The main model I estimate is a local linear regression discontinuity specification:

$$Y_{it} = \alpha + \tau D_{it} + \beta_1 c_{it} + \beta_2 (D_{it} \times c_{it}) + X_{it}' \gamma + \lambda_t + \varepsilon_{it} \tag{3}$$

In this equation, Y_{it} is the outcome for player i drafted in year t. The outcomes include both early-career outcomes and career-level outcomes, such as games played, minutes played, performance measures, and salary. D_{it} is the treatment indicator for being drafted in the first round. The coefficient of interest is τ, which measures the discontinuous change in outcomes at the first-round cutoff. The running variable c_{it} controls for draft position relative to the cutoff, and the interaction term $(D_{it} \times c_{it})$ allows the slope of the running variable to differ on each side of the cutoff. Lastly, X_{it} represents optional player controls, such as height, weight, age at draft, and position, while λ_t represents draft-year fixed effects that absorb differences across draft cohorts.

In the main specification, I use a 10-pick bandwidth around the first-round cutoff, a rectangular kernel that gives equal weight to all players within the bandwidth, and draft-year fixed effects. I do not include additional player controls in the baseline model because the RDD design relies on the comparability of players near the cutoff. As robustness checks, I vary the bandwidth and add player background and pre-draft controls. If the estimated discontinuity remains similar across these specifications, this supports the interpretation that the results are not driven by bandwidth choice, weighting, or observable player characteristics.

This local linear specification follows the standard regression discontinuity setup described in Lee and Lemieux (2010), and Calonico, Cattaneo, and Titiunik (2014) provide data-driven bandwidth selection methods for such designs. Because the running variable is discrete, taking only integer pick positions, I use fixed windows of 5, 10, and 15 picks around the cutoff rather than a data-driven bandwidth. In all specifications, standard errors are cluster-robust on the integer overall pick, following Lee and Card (2008).

The main identifying assumption for the RDD design is that potential outcomes change smoothly around the first-round cutoff. In this context, the assumption means that players drafted just before and just after the cutoff would have had similar expected NBA outcomes if they had received the same draft status. In other words, on average, their underlying abilities in terms of basketball talents should be similar. The only factor that changes discontinuously at the cutoff should be first-round status itself. This assumption would be violated if teams could precisely manipulate which players fall just inside or just outside the first round based on unobserved ability. However, the draft order is determined by team selections, and around the cutoff, the difference between the last first-round pick and the first second-round pick is often minimal. This makes the cutoff a useful setting for comparing similar players who receive different draft labels.

I assess the credibility of the research design through several validity and robustness checks, which are carried out and reported in the results section. First, I check whether pre-draft characteristics, such as height, weight, age, position, and available college statistics, are balanced around the cutoff. If players just inside and just outside the first round look similar before entering the NBA, this supports the assumption that underlying ability changes smoothly at the cutoff. Second, I examine the distribution of observations around the cutoff to make sure there is no unusual bunching or missing data near the boundary. Third, I estimate the model using multiple bandwidths, such as 5-pick, 10-pick, and 15-pick windows around the cutoff, to check whether the results are sensitive to the chosen comparison group. Fourth, I run specifications with and without player controls. If the estimated discontinuity is similar across these models, that would provide stronger evidence that the result is not driven by observable player differences. Additionally, I also use placebo cutoffs away from the actual first-round boundary to check whether similar jumps appear at points where no treatment change occurs.

The methods are appropriate for the research question because the primary objective of this paper is to identify the local average treatment effect of barely receiving first-round status. The RDD directly uses the institutional rule that separates first-round and second-round players. This allows the analysis to move beyond the general fact that higher draft picks perform better and instead focus on whether the first-round label itself creates an additional advantage. Combined with the newly constructed dataset covering draft position, background characteristics, early NBA outcomes, career outcomes, and salary outcomes, this design provides a clear way to study how draft status affects players after they enter the league.

5 Results and Discussion

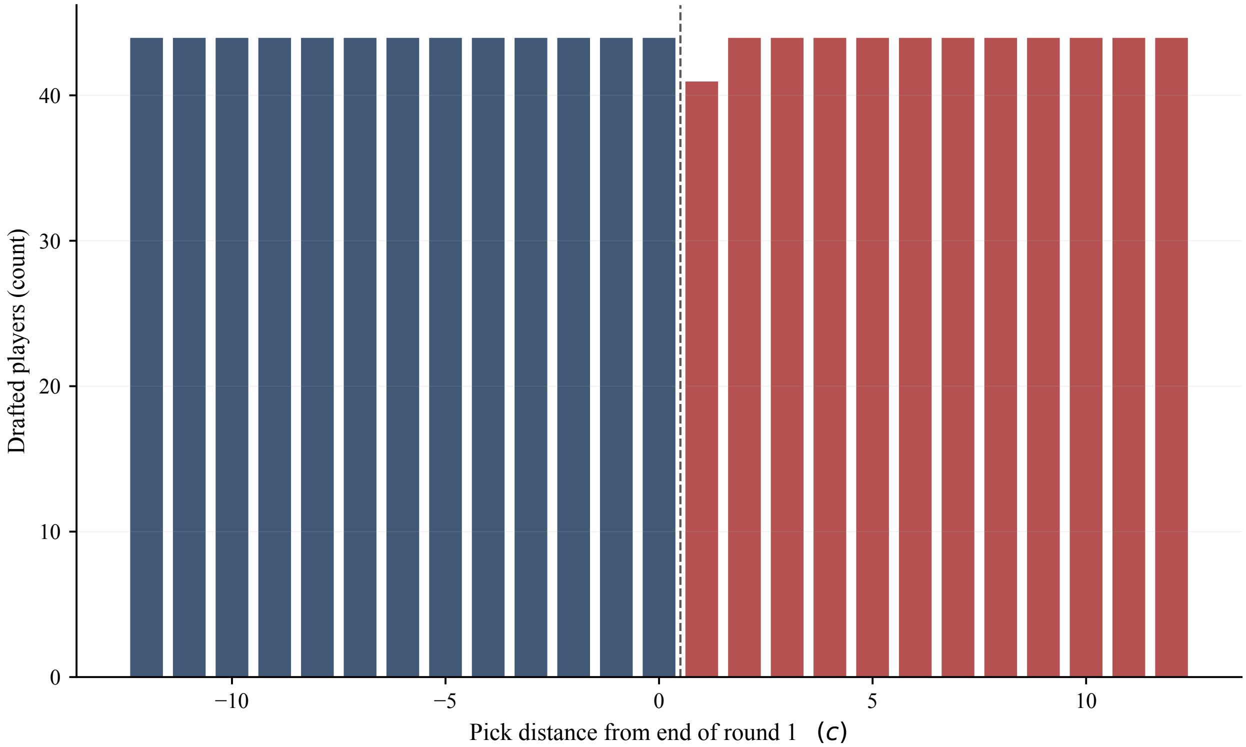
5.1 RDD Validity Checks

Before presenting the main outcome results, I first examine whether the regression discontinuity design is credible in this setting. As discussed in the empirical strategy section, the key identifying assumption is that potential outcomes would have changed smoothly around the first-round cutoff in the absence of first-round status. In this context, players selected just inside and just outside the first-round cutoff should be similar in underlying ability before entering the NBA, and there should not be unusual sorting or missing observations around the cutoff. Although this assumption cannot be tested directly, the data provide several useful checks.

Figure 1 examines the density of the running variable around the cutoff. In this setting, the check is mostly mechanical because the number of picks in each round is set administratively before each draft. Once the round structure is fixed, there is one pick at each draft position, so neither players nor the teams can create bunching just inside or outside the first-round boundary. Therefore, Figure 1 is mainly a formal transparency and data-coverage check. The even distribution confirms that there are no unusual gaps or missing observations around the cutoff. The one exception is $c = +1$, which has 41 rather than 44 observations: in 1997, 2001, and 2002 a forfeited first-round pick was numbered as pick 29 but left unfilled, so the cutoff falls at pick 28, the last player actually selected in the round, and the empty slot appears one position after it.

Figure 2 and Table 2 largely support the balance of pre-draft characteristics around the cutoff. Most observable characteristics are statistically indistinguishable across the boundary, including

Figure 1. Density of running variable around the cutoff (transparency / coverage; pick assignment is administrative)



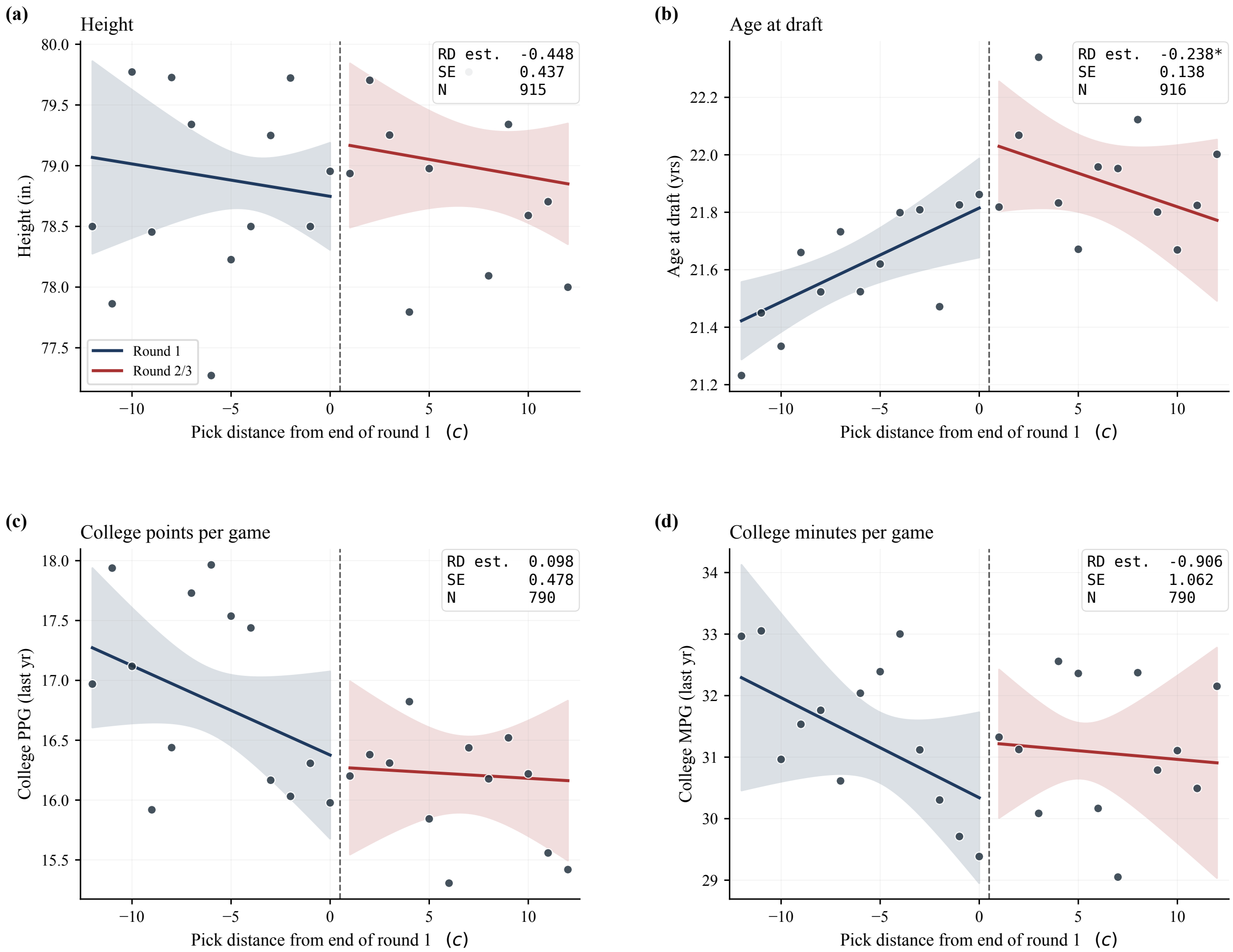
height, weight, position, college games, college minutes, and college points per game.

The main exceptions are age at draft and college field goal percentage. The field goal percentage difference actually works against the main results, since players just outside the first round appear to have slightly higher college shooting efficiency. This makes it less likely that later first-round advantages are simply driven by better observed college performance. The age difference is also small and may reflect how NBA teams value younger players for perceived upside and long-run potential, even when current production is similar. Therefore, while the balance evidence is not perfect, it is mostly consistent with the smoothness assumption. As Table 5 shows, the estimates remain similar after adding controls for height, weight, age at draft, and position, which suggests that the main results are not driven by observable pre-draft differences. Table 5 also reports a specification that adds college field goal percentage and college points per game as controls; the games estimates remain positive and statistically significant.

5.2 Early-Career Outcomes

After establishing the basic credibility of the RDD design, I next examine whether first-round status affects outcomes in the first three NBA seasons. This period is especially important because

Figure 2. Pre-draft characteristics — visual smoothness around the cutoff



Notes: Sharp RDD, local linear, separate slopes, $h = 10$, draft-year FE on y, cluster SE on pick. College FG% is included in Table 2 but not displayed here because it is the only pre-draft variable with a statistically significant jump at the 1 percent level and is discussed in the text as a caveat. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

it captures the initial stage of a player's career, when teams make early decisions about roster spots, playing time, and player development. If the first-round label changes how teams treat otherwise similar players, the effect should appear first in early-career opportunity measures such as games played and minutes played.

Table 3 reports the main first-three-year RDD estimates across bandwidths of 5, 10, and 15 picks around the cutoff. The results show a clear and statistically significant jump in games played. Using the main 10-pick bandwidth, players selected just inside the first round play about 25 more games over their first three seasons than players selected just outside the cutoff. The estimate remains positive and statistically significant across all three bandwidths, suggesting that the result is not driven by a particular choice of comparison window.

The evidence for early-career playing time is also positive, though less robust. At the 10-

Table 2. Balance of pre-draft characteristics across bandwidths (h in {5, 10, 15})

Outcome	$h = 5$	$h = 10$	$h = 15$
Height (in.)	−0.321	−0.448	−0.317
	(0.480)	(0.437)	(0.395)
	N=477	N=915	N=1,352
Weight (lbs)	1.62	−3.27	−1.97
	(3.17)	(2.83)	(2.24)
	N=477	N=915	N=1,352
Age at draft (yrs)	−0.325*	−0.238*	−0.226*
	(0.174)	(0.138)	(0.116)
	N=477	N=916	N=1,353
Position (1=PG . . . 5=C)	0.003	−0.147	−0.072
	(0.199)	(0.204)	(0.168)
	N=458	N=885	N=1,308
College games (last yr)	−1.39	−0.62	−0.67
	(0.95)	(0.73)	(0.60)
	N=412	N=790	N=1,166
College MPG (last yr)	−1.200	−0.906	−0.498
	(1.180)	(1.062)	(0.693)
	N=412	N=790	N=1,166
College PPG (last yr)	−0.465	0.098	−0.202
	(0.756)	(0.478)	(0.416)
	N=412	N=790	N=1,166
College FG% (last yr)	−3.046***	−2.628***	−1.770***
	(1.003)	(0.614)	(0.568)
	N=412	N=790	N=1,166

Notes: Each cell reports the round-1 RD coefficient from a sharp local-linear regression with separate slopes around the year-specific end-of-round-1 cutoff, with draft-year fixed effects on the outcome (Frisch–Waugh) and cluster-robust standard errors on integer overall pick. Rows are pre-draft characteristics; columns vary the bandwidth h. Most coefficients are small and statistically indistinguishable from zero across all bandwidths, supporting the smoothness assumption underlying the RDD. The main exception is college FG%, which shows a small but precisely estimated jump and is discussed in the text. Standard errors in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

pick bandwidth, my main specification, first-round status is associated with roughly 443 additional minutes over the first three seasons, and the estimate is statistically significant at the 10 percent level. The point estimates for points, win shares, and VORP are positive in the main bandwidth but not statistically significant. This pattern suggests that the strongest early effect of first-round

Table 3. First-three-year RDD estimates across bandwidths (h in {5, 10, 15})

Outcome	$h = 5$	$h = 10$	$h = 15$
Games (yrs 1–3)	16.7*	25.4***	22.5***
	(8.7)	(8.2)	(6.5)
	N=480	N=920	N=1,360
Minutes (yrs 1–3)	70	443*	296
	(243)	(260)	(211)
	N=480	N=920	N=1,360
Points (yrs 1–3)	−13	187	129
	(108)	(117)	(98)
	N=480	N=920	N=1,360
Win shares (yrs 1–3)	0.03	0.53	0.18
	(0.80)	(0.73)	(0.61)
	N=480	N=920	N=1,360
VORP (yrs 1–3)	0.023	0.191	0.034
	(0.407)	(0.327)	(0.256)
	N=480	N=920	N=1,360
Salary first 3 yrs ($M)	−0.066	0.118	0.090
	(0.294)	(0.284)	(0.246)
	N=480	N=920	N=1,360

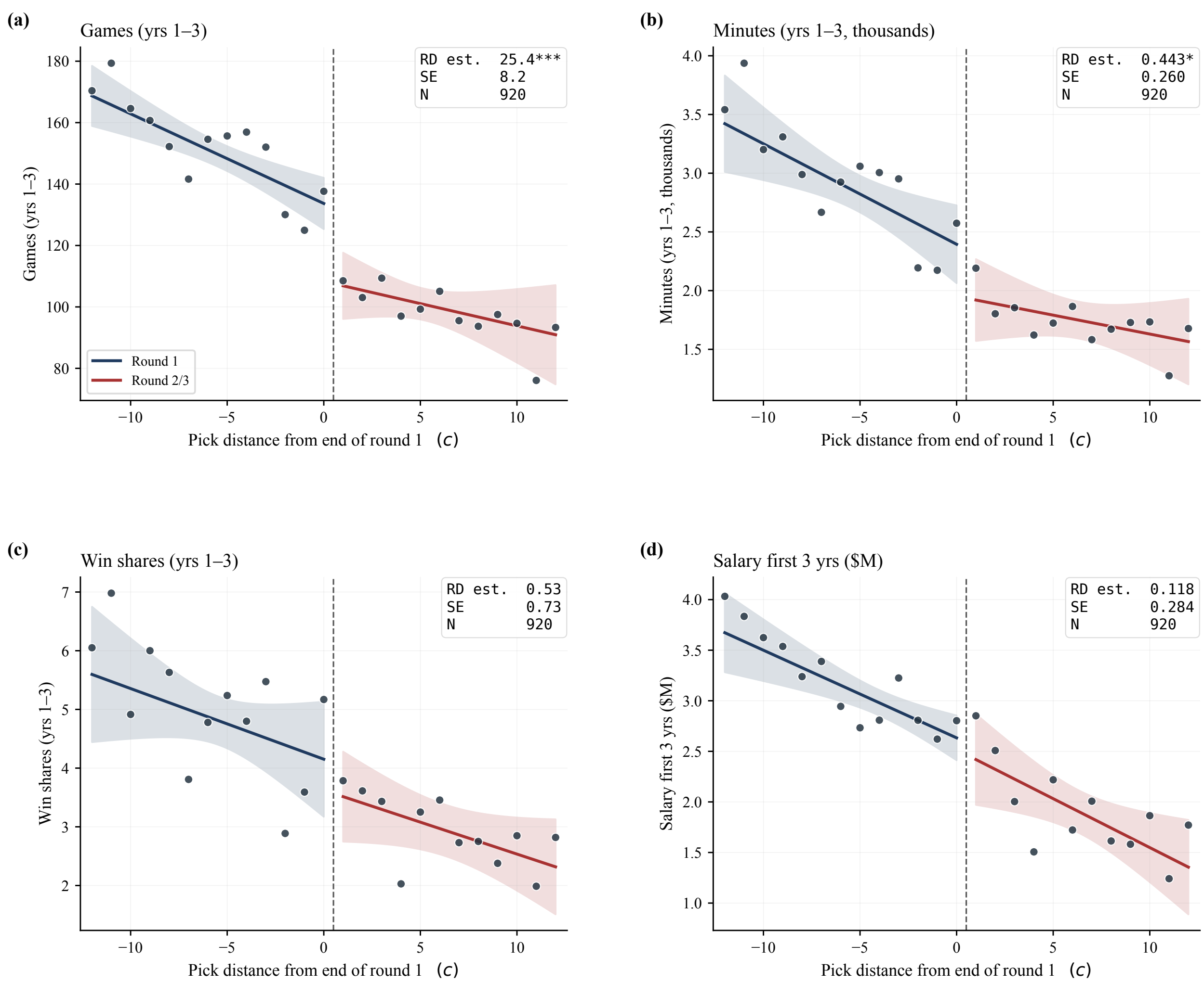
Notes: Sharp local-linear RDD around the year-specific end-of-round-1 cutoff with separate slopes; outcomes are aggregated over each drafted player's first three NBA seasons (intent-to-treat: non-played seasons coded as zero). Draft-year FE on the outcome and cluster-robust SE on integer pick. Round-1 status produces a large, statistically significant gain in games played, a marginally significant gain in minutes at $h = 10$, and positive but statistically insignificant estimates for points, while first-three-year salary shows essentially no jump—consistent with the rookie wage scale compressing initial pay across the cutoff. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

status is on access to playing opportunities rather than clearly higher measured productivity.

The salary result is especially important for interpretation. The estimated jump in first-three-year salary is small and statistically insignificant across all bandwidths. This suggests that the early first-round advantage is not mainly driven by a large immediate salary gap at the cutoff. Instead, the early-career results point more toward roster security and playing-time opportunities: players just inside the first round receive more chances to play, but they do not earn significantly more in their first three seasons.

Figure 3 visualizes some of these patterns. The clearest break at the cutoff appears in games played, with a smaller jump in minutes. Win shares show a positive but imprecise difference, while first-three-year salary is almost completely smooth around the cutoff. Overall, Table 3 and Figure 3 indicate that first-round status creates an early opportunity advantage, but not an immediate salary advantage. This distinction is central to the paper's interpretation because it suggests that the first-round effect mainly operates first through team behavior and playing opportunities, rather than through initial compensation.

Figure 3. Early-career (first three NBA seasons) outcomes around the cutoff



Notes: One observation per drafted player; non-played seasons are zero-imputed for counting outcomes (intent-to-treat). Local linear, separate slopes, $h = 10$, draft-year FE on y, cluster SE on pick. Salary in panel (d) is rookie-scale-driven for first-round picks and shows essentially no jump, consistent with the rookie wage scale compressing initial pay across the cutoff. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

5.3 Career-Level Outcomes

Next, I examine whether the early opportunity advantage translates into longer-run career differences. Table 4 reports RDD estimates for cumulative career outcomes using the main 10-pick bandwidth. Because recent draft classes may still have incomplete careers, the table also reports results after excluding players drafted after 2018, 2016, and 2014. This helps check whether the estimated first-round advantage is driven by younger players who have not yet had enough time to accumulate full career outcomes, or whether the pattern remains among older cohorts with more complete NBA careers.

Table 4. Career RDD estimates by sample restriction ($h = 10$)

Outcome	Full sample	Draft $\leq$ 2018	Draft $\leq$ 2016	Draft $\leq$ 2014
Career length (yrs)	1.27***	1.30**	1.35***	1.40**
	(0.44)	(0.51)	(0.52)	(0.60)
	N=921	N=816	N=774	N=732
Career games	83.5***	83.3**	87.7**	94.9**
	(31.7)	(37.4)	(39.6)	(44.2)
	N=921	N=816	N=774	N=732
Career minutes	2,166**	2,169*	2,202*	2,341*
	(967)	(1,122)	(1,200)	(1,322)
	N=921	N=816	N=774	N=732
Career points	1,034**	974**	1,002**	1,081*
	(415)	(472)	(505)	(558)
	N=921	N=816	N=774	N=732
Career win shares	4.55	4.76	4.81	5.25
	(2.96)	(3.29)	(3.48)	(3.71)
	N=921	N=816	N=774	N=732
Career VORP	1.632	1.758	1.733	1.789
	(1.268)	(1.370)	(1.456)	(1.572)
	N=921	N=816	N=774	N=732
Career salary ($M)	7.17*	7.30*	7.12	6.73
	(3.92)	(4.35)	(4.75)	(5.15)
	N=916	N=811	N=769	N=727

Notes: Sharp local-linear RDD of cumulative career outcomes on round-1 status, $h = 10$, with separate slopes, draft-year FE on the outcome, and cluster-robust SE on integer pick. Recent draft cohorts have right-censored careers; the four columns trim the sample at successively earlier draft years to gauge the sensitivity of the estimates to censoring. Point estimates remain similar across restrictions, indicating that censoring does not drive the round-1 advantage—if anything, restricting to older cohorts widens the estimates. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The career-level results show that first-round status is associated with substantially larger long-run outcomes. In the full sample, players selected just inside the first round play about 1.27 more

NBA seasons, 83.5 more career games, 2,166 more career minutes, and score about 1,034 more career points than players selected just outside the cutoff. These effects remain similar when recent cohorts are removed, suggesting that the estimates are not driven by right-censoring among younger players.

The results for career salary also point in the same direction. In the full sample, first-round status is associated with about $7.17 million more in career earnings, although the estimate is less precise than the playing-time outcomes. By contrast, value-based measures such as career win shares and VORP are positive but not statistically significant. This pattern suggests that the strongest evidence is for differences in career opportunity and cumulative production, rather than a clear discontinuity in underlying productivity or efficiency, which further supports the validity of the RDD research design.

Figure 4 provides visual support for the above findings. The largest jump appears in career games, while career win shares, VORP, and salary also show positive differences at the cutoff but with wider confidence intervals. The career-level evidence suggests that the early first-round advantage does not disappear after the first few seasons. Instead, greater early opportunity appears to accumulate into longer careers, more total playing time, higher cumulative production, and larger career earnings.

5.4 Season-by-Season Persistence

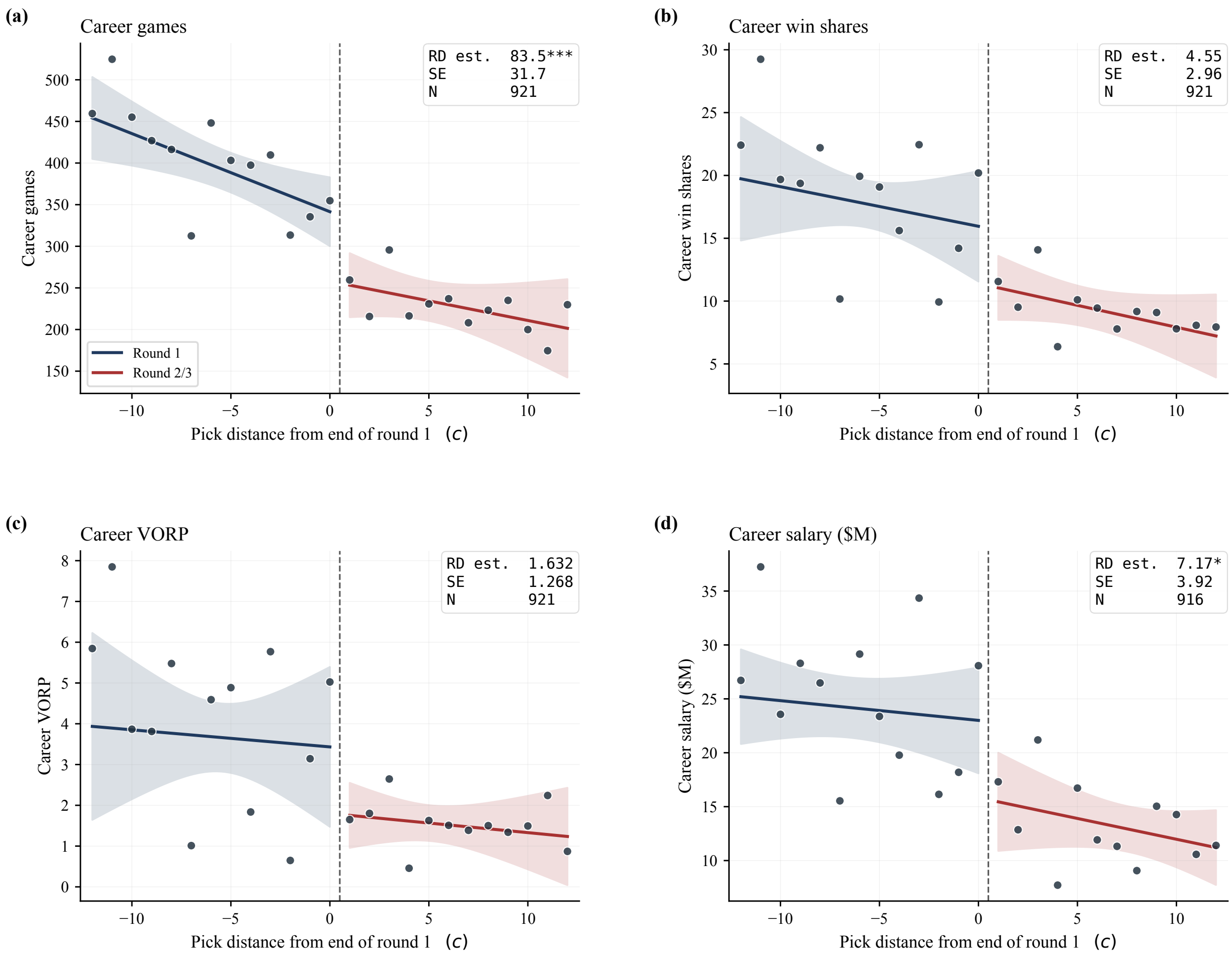
The next question is whether the first-round advantage is concentrated in the first few seasons or continues into later stages of a player's career. Figure 5 reports separate RDD estimates by career year for games, minutes, points, win shares, salary, and box plus/minus. The vertical dotted line marks the end of the fourth season, which is approximately when the initial rookie-scale contract period ends for first-round picks.

The figure shows that the first-round effect is generally positive across career years, especially for games and minutes. The estimates are largest in the early seasons, but they remain mostly above zero even after the first four years. This suggests that the first-round advantage is not limited to the initial contract period. Instead, early differences in playing time and roster security appear to carry forward into later career opportunities.

The evidence is weaker for performance-rate measures. Points and win shares show mostly positive but less precise estimates, while box plus/minus is noisy and does not show a consistent pattern. This reinforces the earlier interpretation that the first-round effect is strongest for opportunity-based outcomes rather than clear differences in underlying efficiency.

Overall, Figure 5 supports the idea that first-round status can shape a player's career path beyond the first few seasons. The advantage begins early, but it does not fully disappear once

Figure 4. Career-level outcomes around the cutoff (full sample 1980–2023)



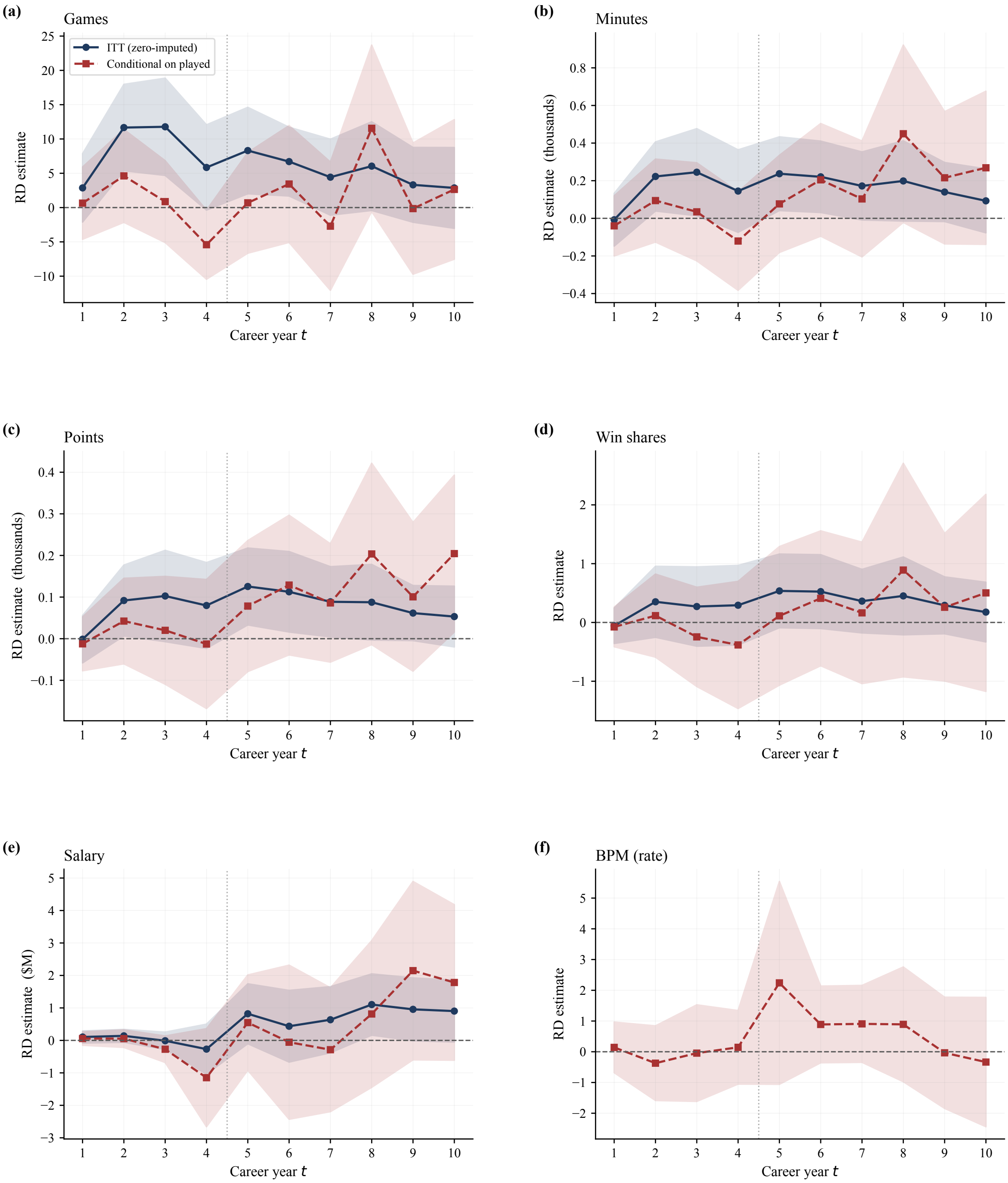
Notes: One observation per drafted player. Non-played career totals are zero-imputed (intent-to-treat). Local linear, separate slopes, $h = 10$, draft-year FE on y, cluster SE on pick. Recent cohorts (draft year > 2016) have right-censored careers; Table 4 reports both the full sample and the draft year $\leq$ 2016 subsample. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

rookie contracts expire, suggesting that early organizational investment may have lasting effects on NBA career development.

5.5 Robustness Checks and Era Splits

Finally, I investigate whether the main results are robust to alternative specifications and whether the first-round effect differs across draft eras. Table 5 reports estimates using different bandwidths and a specification with pre-draft player controls. The main opportunity-based results are stable across these checks. Career length, career games, and first-three-year games remain positive and statistically significant across the 5-pick, 10-pick, and 15-pick bandwidths. The estimates also remain similar after adding controls for height, weight, age at draft, and position,

Figure 5. Persistence of the round-1 effect by career year



Notes: Each point is a sharp local-linear RDD estimate of the first-round effect for the outcome measured in career year t, bandwidth $h = 10$, draft-year FE on y, cluster-robust SE on pick. ITT (zero-imputed) treats non-played seasons as zero; conditional restricts to seasons actually played. BPM is a rate, so only the conditional series is shown. Vertical dotted line at $t = 4.5$ marks the end of the four-year rookie-scale contract for first-round picks. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

which suggests that the main results are not driven by observable pre-draft differences. Appendix D reports the same estimates with heteroskedasticity-robust standard errors in place of clustering on pick; the full-sample playing-time estimates remain significant at the 5 percent level.

Table 5. Robustness — bandwidth and controls (h in {5, 10, 15}, year FE always included)

Outcome	$h = 5$	$h = 10$	$h = 15$	$h = 10$ + controls	$h = 10$ + controls + college
Career length	1.07**	1.27***	1.20***	1.14**	1.47***
	(0.52)	(0.44)	(0.40)	(0.48)	(0.47)
	N=481	N=921	N=1,361	N=885	N=765
Career games	69.0**	83.5***	73.1**	69.8**	89.1**
	(32.6)	(31.7)	(29.6)	(34.1)	(35.3)
	N=481	N=921	N=1,361	N=885	N=765
Career win shares	3.88	4.55	2.54	2.87	3.69
	(3.82)	(2.96)	(2.64)	(2.95)	(3.10)
	N=481	N=921	N=1,361	N=885	N=765
Career salary ($M)	6.34	7.17*	4.90	3.61	5.31
	(5.35)	(3.92)	(3.66)	(3.79)	(3.74)
	N=477	N=916	N=1,353	N=885	N=765
First-3 games	16.7*	25.4***	22.5***	23.3***	32.7***
	(8.7)	(8.2)	(6.5)	(8.3)	(8.6)
	N=480	N=920	N=1,360	N=885	N=765
First-3 salary ($M)	−0.07	0.12	0.09	0.17	0.44**
	(0.29)	(0.28)	(0.25)	(0.29)	(0.19)
	N=480	N=920	N=1,360	N=885	N=765

Notes: Sharp local-linear RDD with separate slopes around the year-specific end-of-round-1 cutoff. The first three columns vary the bandwidth h; the fourth column adds pre-draft controls (height, weight, age at draft, position) to the $h =$ 10 specification. The fifth column further adds college field goal percentage and college points per game from the player's last NCAA season; the sample is smaller because college statistics are unavailable for some international and high school players. All columns include draft-year FE on the outcome (Frisch–Waugh) and cluster-robust SE on integer pick. The round-1 effect on career length, career games, and first-three-year games is stable across all bandwidths and is essentially unchanged by adding controls; salary point estimates attenuate when the pre-draft controls are included and rise in the college-controls column, though they remain small relative to the playing-time effects. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The salary results are less stable. Career salary is positive in all bandwidths but statistically significant only at the 10-pick bandwidth, and it becomes smaller and insignificant after adding controls. First-three-year salary remains small and statistically insignificant in every specification except the college-controls column, where it is positive and significant at the 5 percent level but economically small at about $0.44 million over three seasons. This pattern reinforces the earlier finding that the first-round effect is clearer for opportunity-based outcomes than for pay outcomes, and shows that this conclusion is not sensitive to bandwidth choice or the inclusion of observable player controls.

Table 6 and Figure 6 examine whether the effect differs across draft eras. This split is important because the institutional meaning of first-round status changed over time, especially after the introduction of the modern rookie wage scale in 1995. The largest effects generally appear in the 1995–2004 period, when first-round status was tied to stronger contractual guarantees. In that era, first-round status is associated with larger and statistically significant jumps in NBA entry, career length, career games, and career win shares. The relevant CBA changes and their potential relationship to the first-round discontinuity are discussed in greater detail in Appendix B.

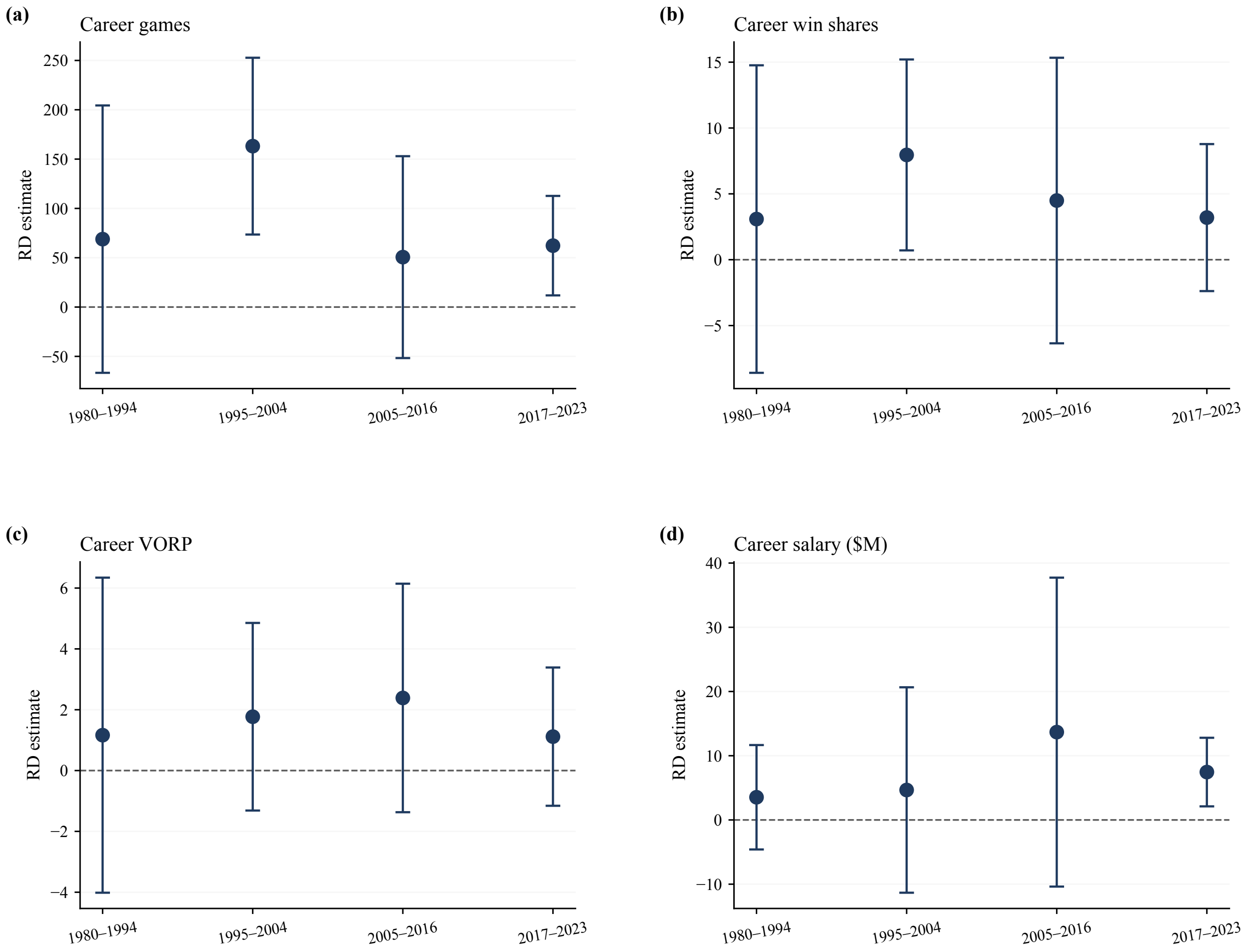
Table 6. CBA / draft-era split ($h = 10$, year FE within era)

Outcome	1980–1994	1995–2004	2005–2016	2017–2023
Pr(NBA)	0.060	0.151***	−0.062*	0.076
	(0.078)	(0.049)	(0.037)	(0.049)
	N=315	N=207	N=252	N=147
Career yrs	0.99	2.55***	0.84	0.83**
	(0.98)	(0.61)	(0.67)	(0.42)
	N=315	N=207	N=252	N=147
Career G	68.9	163.1***	50.6	62.3**
	(69.2)	(45.7)	(52.2)	(25.7)
	N=315	N=207	N=252	N=147
Career WS	3.09	7.96**	4.49	3.20
	(5.96)	(3.70)	(5.53)	(2.85)
	N=315	N=207	N=252	N=147
Career salary ($M)	3.53	4.66	13.68	7.46***
	(4.15)	(8.16)	(12.27)	(2.72)
	N=314	N=205	N=250	N=147

Notes: Sample split by draft year of the pick into four CBA / draft regimes. Sharp local-linear RDD within each era, $h = 10$, with separate slopes, draft-year FE within the era, and cluster-robust SE on integer pick. Pre-1995 cohorts predate the modern rookie wage scale and show smaller, statistically insignificant effects, consistent with the absence of a contractual reason to treat round-1 and round-2 picks differently. The 1995–2004 era — when the rookie wage scale and guaranteed first-round contracts were introduced — shows the largest round-1 jumps in playing-time and probability of playing in the NBA. Effects attenuate after the 2005 CBA reduced the number of guaranteed first-round years. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The effects became smaller and less precise after 2005, consistent with Zhou et al.'s finding that the 2005 CBA reduced guaranteed years for first-round contracts and weakened the first-round premium. However, the estimates are generally positive in all eras, including the pre-1995 and post-2017 periods. This signals that the first-round advantage is not purely mechanical or entirely

Figure 6. Round-1 RDD treatment effects by CBA / draft era



Notes: Sample split by draft year of the pick. Each point is the sharp RD estimate $\hat{\tau}$ with bandwidth $h = 10$ and draft-year FE within the regime; bars are 95% CI from cluster-robust SE on pick. Pre-1995 cohorts predate the modern rookie wage scale, so contract-driven mechanisms need not align with post-1995 eras.

explained by one CBA rule. Instead, the evidence points to a broader mechanism in which the first-round label affects team investment, roster security, and playing opportunities, while contract rules strengthen or weaken that effect across eras. The one exception is the probability of NBA entry in the 2005–2016 period, which is negative and marginally significant, although the career outcome estimates in that era remain positive.

5.6 Placebo Tests

As a final robustness check, I estimate placebo discontinuities at fake cutoffs where no actual change in draft status occurs. The purpose of this test is to examine whether the main results are specific to the true first-round boundary, or whether similar jumps appear at arbitrary points within the first or second round.

Table 7 compares the real cutoff at $c = 0$ with placebo cutoffs within the same round. The placebo at $c = -5$ is located within the first round, while the placebo cutoffs at $c = +5$ and $c = +10$ are located within the second round. For these placebo tests, I use a 5-pick bandwidth so that the comparison window does not cross the true first-round cutoff.

Table 7. Placebo cutoffs — real cutoff vs fake cutoffs within round ($h = 5$ for placebos to avoid spanning the real cutoff)

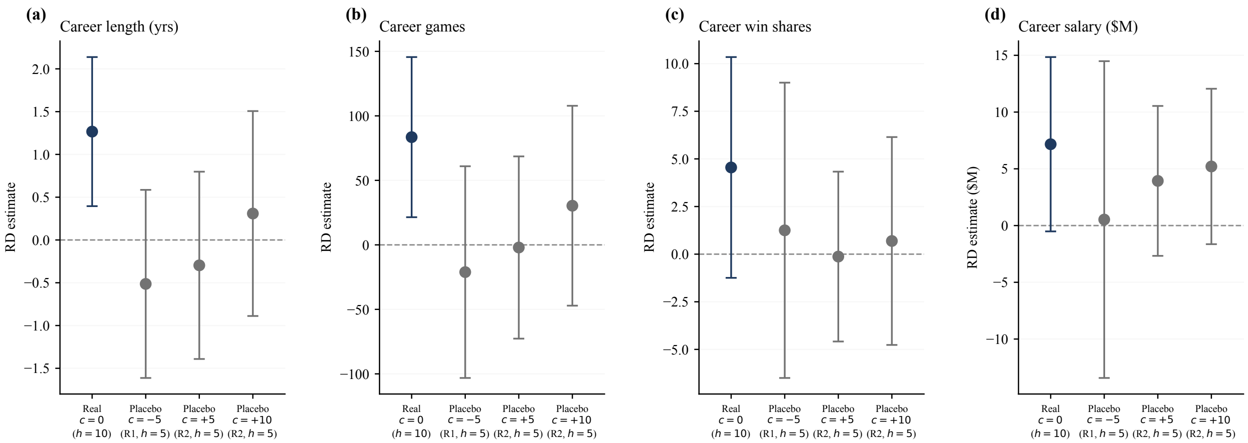
Outcome	Real cutoff $c = 0$	Placebo $c = -5$ (within R1, $h = 5$)	Placebo $c = +5$ (within R2, $h = 5$)	Placebo $c = +10$ (within R2, $h = 5$)
Career length	1.27***	−0.51	−0.30	0.31
	(0.44)	(0.56)	(0.56)	(0.61)
	N=921	N=484	N=437	N=484
Career games	83.5***	−21.1	−2.0	30.3
	(31.7)	(41.9)	(36.0)	(39.5)
	N=921	N=484	N=437	N=484
Career win shares	4.55	1.25	−0.13	0.69
	(2.96)	(3.95)	(2.27)	(2.78)
	N=921	N=484	N=437	N=484
Career salary ($M)	7.17*	0.53	3.93	5.21
	(3.92)	(7.12)	(3.37)	(3.49)
	N=916	N=483	N=433	N=480

Notes: Placebo specifications check whether spurious jumps appear at fake cutoffs where no treatment change occurs. Each placebo column restricts the sample to a single round (round 1 for $c = -5$, round 2 for $c = +5$ and $c = +10$) so that the bandwidth window does not span the real cutoff at $c = 0$. Bandwidth $h = 5$ picks; otherwise the specification matches the headline RDD (separate slopes, draft-year FE on the outcome, cluster-robust SE on integer pick). The first column reproduces the real-cutoff estimate. The placebo coefficients are small and statistically insignificant for every outcome, supporting the interpretation that the round-1 jumps documented in Tables 3–4 reflect the institutional cutoff rather than a smooth underlying gradient in pick distance. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The results provide additional support for the main interpretation. At the real cutoff, the estimates are positive and statistically significant for career length, career games, and career salary. In contrast, the placebo estimates are smaller and statistically insignificant for every outcome. For example, the career games effect is 83.5 games at the true cutoff, but the placebo estimates are −21.1, −2.0, and 30.3 games, with larger variations at the fake cutoffs. A similar pattern appears for career length and win shares.

Figure 7 shows the same comparison visually. The real cutoff produces a clear positive estimate for career length and career games, while the placebo cutoffs have confidence intervals that include zero. Overall, the placebo tests suggest that the estimated first-round effects are not simply capturing a smooth relationship between draft position and career outcomes. Instead, the jumps appear concentrated at the actual institutional boundary between the first and second rounds.

Figure 7. Placebo cutoffs — RDD estimates at the real cutoff vs three within-round fake cutoffs



Notes: Round-1 RD estimate (point) with 95% CI from cluster-robust SE on integer pick. Blue: real cutoff (full sample, $h = 10$). Gray: placebo cutoffs restricted to a single round so the bandwidth window does not span the real cutoff ($h = 5$). Specifications match Table 7. Placebo estimates are small and statistically indistinguishable from zero, supporting the institutional-cutoff interpretation. Draft-year FE on y throughout.

6 Conclusions

This paper investigates the causal effect of being selected in the first round of the NBA Draft on a player's subsequent NBA career outcomes, using a regression discontinuity design centered on the institutional boundary between the first and second rounds. The central motivation is that two players drafted one pick apart, one with first-round status and one without, may face drastically different early-career treatments such as organizational expectations, contract guarantees, and development opportunities, even though the differences in their underlying talent are nearly indistinguishable. Using a player-level dataset constructed from Basketball-Reference covering all picks within the first 60 selections of every draft from 1980 to 2023, the paper asks whether this label itself, independent of ability, shapes career outcomes.

The main findings are consistent and robust across specifications. Players selected just inside the first round play approximately 25 more games and 443 more minutes over their first three seasons than comparable players selected just outside the cutoff. These early-career differences then compound: first-round players go on to play roughly 1.3 more NBA seasons, appear in about 84 more career games, accumulate over 2,100 more career minutes, and score approximately 1,000 more career points. A career earnings advantage of around $7 million is also observed, though this result is somewhat less precise from the statistical standpoint. Crucially, salary in the first three years shows essentially no jump at the cutoff, which is consistent with the rookie wage scale compressing initial pay across the first-round cutoff. This rules out immediate compensation as the primary channel and instead points toward roster security, organizational patience, and early playing opportunities as the mechanisms through which first-round status compounds into long-run career differences. The season-by-season analysis further shows that this gap does not dissolve once rookie contracts expire, suggesting that early opportunity advantages translate into durable career trajectories.

Though the research designs and objectives are quite similar, this paper builds on and extends Zhou, Hu, and Shi (2023) in four meaningful ways. First, the sample is expanded from the 1995 to 2019 window to cover 1980 to 2023, spanning the pre-rookie-scale era, multiple CBA reforms, and more recent draft cohorts, enabling direct estimation of how the first-round premium tracks institutional changes in contract rules. Second, early-career outcomes are separated from long-run career outcomes, rather than collapsed into a single cumulative measure. This separation reveals that the first-round advantage operates primarily through opportunity accumulation rather than an immediate salary premium, a distinction that is obscured when only cumulative career outcomes are examined as a single aggregate measure. Third, a broader set of value-based performance metrics including win shares, VORP, and box plus/minus is incorporated alongside cumulative production measures, enabling an assessment of whether the discontinuity reflects differences in productivity or differences in opportunity. The evidence consistently favors opportunity: value-based measures are generally positive but imprecise, while playing-time outcomes are large and statistically significant. Fourth, the season-by-season persistence analysis directly tests whether the first-round gap is concentrated in the rookie contract period or continues beyond it. The finding that the gap persists past year four is difficult to reconcile with a purely mechanical contract story and instead points toward a broader organizational investment and reputation mechanism.

Several limitations are relevant to these findings. The most important is the absence of private scouting information. The validity of the RDD rests on the assumption that underlying player ability changes smoothly at the first-round cutoff, and while pre-draft observable characteristics are largely balanced around the round-1 cutoff, teams hold private information including workout results, medical reports, interview evaluations, and internal draft boards that cannot be observed in public data. If teams systematically use such information to distinguish between players right at the boundary, the identifying assumption would be weakened or violated. Furthermore, even if the RDD assumptions are satisfied, the design identifies a local average treatment effect for players near the first-round cutoff, not the average effect of first-round status for all first-round picks. Therefore, the results should not be generalized to lottery picks or players selected much earlier in the draft. Additionally, salary data are incomplete for a portion of the earlier cohorts, and career outcomes for players drafted after the late 2010s are right-censored, limiting the precision of long-run estimates for more recent draft classes.

The ideal analysis would address these limitations through access to private scouting data. If one could obtain teams' internal pre-draft rankings or composite scouting grades for every player in each draft class, these could serve as direct controls for unobserved ability and sharpen the validity of the RDD. Such data would also allow a more precise test of whether the identifying assumption holds, specifically whether scouting grades change smoothly at the first-round cutoff or display their own discontinuity. Beyond data improvements, a natural extension would be to trace the mechanism more directly by examining within-team decisions: whether coaches allocate more minutes to first-round players conditional on similar early season performance, and whether training staff investments and contract extension decisions differ at the cutoff. Such within-team variation would allow a cleaner separation of the sunk-cost and organizational-investment channels, which both this paper and Zhou et al. identify as important but cannot fully disentangle with the available data. Additionally, I would investigate treatment heterogeneity, which would be especially interesting for understanding whether the first-round effect differs across groups such as college players, international players, guards, wings, and bigs. A further extension would collect richer contract, injury, G League, and transaction data to better trace whether the first-round advantage comes through guaranteed contracts, roster security, development opportunities, or later career mobility. Together, these extensions would move the analysis from documenting that the first-round label matters toward understanding precisely how and through whom that advantage is conferred. However, the evidence in this paper still suggests that draft labels can meaningfully shape opportunity, investment, and long-run career development.

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APPENDIX

Appendix A. Overview of the NBA Draft Mechanism

The NBA Draft is the primary process through which NBA teams acquire the rights to incoming players. In the modern format, the draft consists of two rounds, with each team generally holding one pick per round before trades, forfeitures, or other adjustments. The draft has used a two-round format since 1989; before that, the draft included more rounds, including three rounds in 1988 and seven rounds in 1987 (Sporting News, 2024). Because the number of NBA teams has changed over time, the total number of picks in each round has also changed across draft years.

The first round is partly determined by the NBA draft lottery. The lottery was introduced in 1985 to determine the order of the top picks among non-playoff teams. In the modern system, lottery odds are based on regular-season records, with worse teams generally receiving higher odds of obtaining earlier picks. After the lottery positions are assigned, the rest of the first round and the second round generally follow reverse order of regular-season record, subject to trades and forfeited picks (National Basketball Association, 2025).

Draft picks can be traded before or during the draft, so the team making a selection is not always the team originally assigned that pick. Teams may also lose picks because of league penalties, which can change the total number of selections in a given draft. For these reasons, the exact final pick of the first round can vary across years. In this paper, a player's round status is therefore determined using the actual round structure in that draft year rather than a fixed overall pick number.

Appendix B. CBA Changes and Era Classification

The contractual meaning of first-round status changed during the sample period. The key change was the 1995 Collective Bargaining Agreement, which introduced the modern rookie wage scale for first-round picks. Under the 1995 system, first-round picks received three guaranteed contract years with salaries set by a rookie scale, while second-round picks were not governed by the same standardized structure (Groothuis, Hill, & Perri, 2007). This is the main reason this paper separates the pre-1995 era from later draft years.

The paper uses four CBA-era groups: 1980–1994, 1995–2004, 2005–2016, and 2017–2023. The pre-1995 period predates the modern rookie wage scale. The 1995–2004 period captures the initial rookie-scale era, when first-round picks had three guaranteed years. The 2005–2016 period follows the 2005 CBA change, which reduced rookie-scale guarantees from three guaranteed seasons to two guaranteed seasons with team options for the third and fourth years (DraftExpress, 2005; Zhou, Hu, & Shi, 2023). The 2017–2023 period follows the 2017 CBA, which increased the rookie salary scale by 45 percent over a three-year phase-in while preserving the basic first-round rookie-scale structure (NBC Sports, 2017). In this paper, these era splits are used to examine whether the first-round discontinuity becomes larger or smaller when the contractual value of first-round status changes.

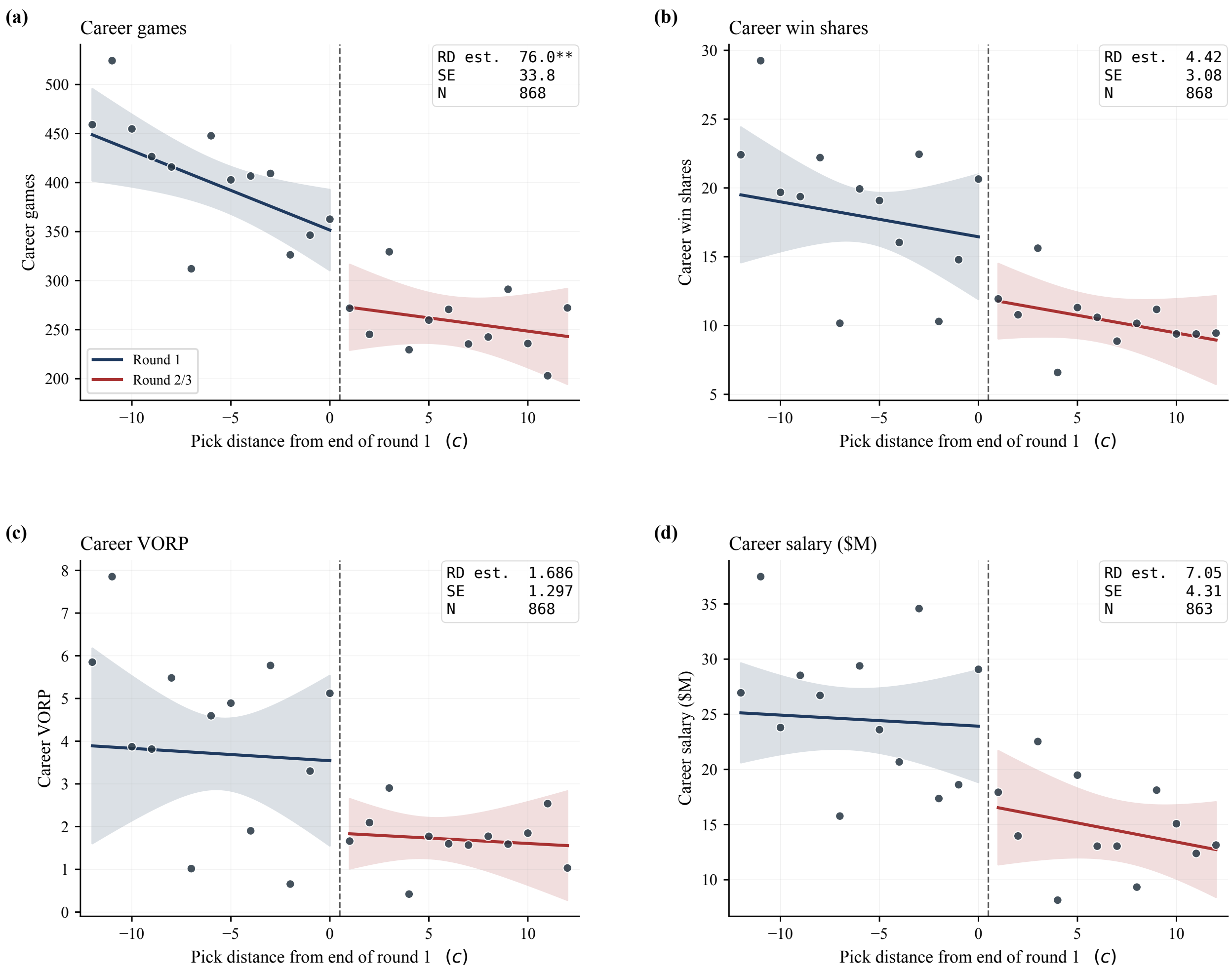
Appendix C. Intensive-Margin Career Outcomes (Additional Robustness Check)

As an additional robustness check, Appendix Figure A1 and Appendix Table A1 restrict the sample to players who appeared in at least one NBA game. This specification removes players who were drafted but never played in the NBA, allowing the analysis to test whether the round-1 premium is driven only by the extensive margin of reaching the league. The results remain broadly similar: first-round status is still associated with longer careers, more career games, more minutes, and more points among players who appear in the NBA. The estimates are somewhat smaller and less precise after adding controls, but the overall pattern suggests that the first-round advantage operates primarily through intensive-margin differences in opportunity and career development, not only through whether a player reaches the NBA at all.

Appendix D. Inference Without Clustering

Because the running variable takes only integer values, cluster-robust standard errors on the running variable can be unreliable when the number of clusters is small. Kolesár and Rothe (2018) show that confidence intervals based on this approach can undercover when the running variable is discrete, which motivates the comparison below. As a check, Appendix Table A2 re-estimates Table 4 with heteroskedasticity-robust (HC1) standard errors in place of clustering on pick. Point estimates are unchanged by construction. In the full sample, the estimates for career length, career games, career minutes, and career points remain significant at the 5 percent level under HC1; nine of the 28 cells change star level, and no estimate changes sign.

Figure A1. Career outcomes around the cutoff — conditional on appearing in the NBA



Notes: Mirrors Figure 4's panels but restricts the sample to players who appeared in at least one NBA game (played_nba = 1). Bin-scatter mean at each integer pick distance c with round-1 (blue) and round-2/3 (red) local-linear fits and pointwise 95% confidence bands. Comparing against Figure 4, the round-1 jumps remain large and visually similar once never-NBA picks are removed, indicating that the round-1 premium operates primarily through the intensive margin — differences in career length, playing time, and earnings among players who reach the NBA — rather than through the extensive margin alone. $h = 10$, draft-year FE on y, cluster-robust SE on integer pick. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table A1. Round-1 RDD on career outcomes — full sample (ITT) vs conditional on appearing in the NBA

Outcome	Full sample $h = 10$	Full sample $h = 10$ + ctrl	Played NBA only $h = 10$	Played NBA only $h = 10$ + ctrl
Career length (yrs)	1.27***	1.14**	1.09**	0.99**
	(0.44)	(0.48)	(0.45)	(0.47)
	N=921	N=885	N=868	N=838
Career games	83.5***	69.8**	76.0**	63.5*
	(31.7)	(34.1)	(33.8)	(35.7)
	N=921	N=885	N=868	N=838
Career minutes	2,166**	1,562	2,074**	1,489
	(967)	(1,023)	(1,027)	(1,065)
	N=921	N=885	N=868	N=838
Career points	1,034**	668	1,021**	663
	(415)	(437)	(429)	(444)
	N=921	N=885	N=868	N=838
Career win shares	4.55	2.87	4.42	2.77
	(2.96)	(2.95)	(3.08)	(3.04)
	N=921	N=885	N=868	N=838
Career VORP	1.632	0.890	1.686	0.949
	(1.268)	(1.182)	(1.297)	(1.199)
	N=921	N=885	N=868	N=838
Career salary ($M)	7.17*	3.61	7.05	3.64
	(3.92)	(3.79)	(4.31)	(4.13)
	N=916	N=885	N=863	N=838

Notes: Sharp local-linear RDD with separate slopes around the year-specific end-of-round-1 cutoff, $h = 10$, draft-year FE on the outcome (Frisch–Waugh), and cluster-robust SE on integer pick. The first two columns are the intent-to-treat (ITT) specification used in the main results: every drafted player is included and never-NBA picks contribute zeros for cumulative outcomes. The third and fourth columns restrict to players with at least one NBA appearance (played_nba = 1), isolating intensive-margin differences in career outcomes among players who reached the league. Pre-draft controls (height, weight, age at draft, position) appear in columns 2 and 4. Comparing column 1 with column 3, roughly 85–100% of the ITT round-1 effect persists after dropping never-NBA picks, indicating that the cumulative round-1 premium is driven primarily by the intensive margin — differences in career length, playing time, and earnings among players who do reach the NBA — rather than by selection into the league. Adding pre-draft controls (columns 2 and 4) attenuates the salary point estimates, consistent with the rookie wage scale tying first-round salary to draft position itself, but leaves the playing-time estimates broadly unchanged. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A2. Career RDD estimates ($h = 10$): cluster-robust versus HC1 standard errors

Outcome	Full sample	Draft $\leq$ 2018	Draft $\leq$ 2016	Draft $\leq$ 2014
Career length (yrs)	1.27	1.30	1.35	1.40
Cluster-robust SE	(0.44)***	(0.51)**	(0.52)***	(0.60)**
HC1 SE	(0.55)**	(0.62)**	(0.64)**	(0.67)**
	N=921	N=816	N=774	N=732
Career games	83.5	83.3	87.7	94.9
Cluster-robust SE	(31.7)***	(37.4)**	(39.6)**	(44.2)**
HC1 SE	(38.1)**	(42.6)*	(44.5)**	(46.3)**
	N=921	N=816	N=774	N=732
Career minutes	2,166	2,169	2,202	2,341
Cluster-robust SE	(967)**	(1,122)*	(1,200)*	(1,322)*
HC1 SE	(1,040)**	(1,162)*	(1,210)*	(1,260)*
	N=921	N=816	N=774	N=732
Career points	1,034	974	1,002	1,081
Cluster-robust SE	(415)**	(472)**	(505)**	(558)*
HC1 SE	(449)**	(500)*	(518)*	(536)**
	N=921	N=816	N=774	N=732
Career win shares	4.55	4.76	4.81	5.25
Cluster-robust SE	(2.96)	(3.29)	(3.48)	(3.71)
HC1 SE	(2.82)	(3.16)	(3.29)	(3.41)
	N=921	N=816	N=774	N=732
Career VORP	1.632	1.758	1.733	1.789
Cluster-robust SE	(1.268)	(1.370)	(1.456)	(1.572)
HC1 SE	(0.969)*	(1.085)	(1.125)	(1.173)
	N=921	N=816	N=774	N=732
Career salary ($M)	7.17	7.30	7.12	6.73
Cluster-robust SE	(3.92)*	(4.35)*	(4.75)	(5.15)
HC1 SE	(4.28)*	(4.81)	(4.99)	(5.12)
	N=916	N=811	N=769	N=727

Notes: Same specification, samples, and outcomes as Table 4 (sharp local-linear RDD of cumulative career outcomes on round-1 status, $h = 10$, with separate slopes and draft-year FE on the outcome). Point estimates are identical by construction. For each outcome, the first row of standard errors is cluster-robust on integer overall pick (28 clusters at $h = 10$), following Lee and Card (2008); the second row is heteroskedasticity-robust (HC1) without clustering. Stars next to each standard error refer to that standard error. Nine of the 28 cells change star level and no estimate changes sign. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.